Voyager Spacecraft RF Link Verification Report

Automated Software Test Results

Item	Result
Test command	python -m pytest -q
Overall test status	PASS
Pytest summary	16 passed in 0.77s
Process exit code	0

Educational RF communications simulation

Generated: 2026-11-04 16:49:13 EST
Measurement source: Simulated calculations. No physical RF instruments or live spacecraft telemetry were connected.

1. Test Configuration

Parameter	Value
Carrier-to-noise density ratio (C/N0)	31.01 dB-Hz
Minimum Eb/N0 criterion	3.00 dB
Criterion source	Educational example; not an official NASA requirement

2. Verification Results

Scenario	Data Rate (bps)	Eb/N0 (dB)	Minimum (dB)	Margin (dB)	Result
Normal real-time telemetry	160	8.97	3.00	+5.97	PASS
High-rate playback	1,400	-0.45	3.00	-3.45	FAIL
Historical high-rate capability	7,200	-7.56	3.00	-10.56	FAIL

Interpretation: a non-negative margin meets the configured educational criterion. These results do not establish actual Voyager operating capability.

3. Requirements Traceability

Requirement ID	Requirement	Method	Test File / Function	Expected Result	Status
RF-VER-001	Calculate Eb/N0 correctly for a known input	Unit test	test_verification.py / test_calculate_eb_n0_at_160_bps	Calculated value matches expected result	Implemented
RF-VER-002	Confirm Eb/N0 decreases when data rate increases at fixed C/N0	Unit test	test_verification.py / test_calculate_eb_n0_decreases_as_data_rate_increases	Higher data rate produces lower Eb/N0	Implemented
RF-VER-003	Pass when calculated link margin is positive	Unit test	test_verification.py / test_link_passes_when_margin_is_positive	Verification returns PASS	Implemented
RF-VER-004	Fail when calculated link margin is negative	Unit test	test_verification.py / test_link_fails_when_margin_is_negative	Verification returns FAIL	Implemented
RF-VER-005	Pass when calculated link margin is exactly zero	Boundary-value test	test_verification.py / test_link_passes_when_margin_is_exactly_zero	Verification returns PASS	Implemented
RF-VER-006	Reject a zero telemetry data rate	Invalid-input test	test_verification.py / test_zero_data_rate_is_rejected	ValueError is raised	Implemented
RF-VER-007	Reject a negative telemetry data rate	Invalid-input test	test_verification.py / test_negative_data_rate_is_rejected	ValueError is raised	Implemented

4. Limitations

- C/N0 is a fixed model input, not a live measurement.

- The minimum Eb/N0 criterion is illustrative, not a NASA requirement.

- Actual link performance depends on modulation, coding, receiver characteristics, system losses, and operating conditions.

- Traceability statuses are read from the CSV and are not independently confirmed by this report generator.

End of report.